ELEC70098
EE9-CS3-2
EE9-FPN1-04

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

OPTIMISATION

Thursday, 1 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) :

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OPTIMISATION

1. Consider the (continuous) function

$$f(x,y) = xy(x+y-1).$$

a) By using first order necessary conditions of optimality determine all stationary points of the function f.
(Hint: the function has 4 stationary points.) [4 marks]

b) By using second order sufficient conditions of optimality classify the stationary points of the function f determined in part a). [4 marks]

c) By evaluating the function along the line $y = x$ show that the function is not bounded from below, nor bounded from above, hence it does not have a global minimizer nor a global maximizer. [2 marks]

d) Write the Gradient iteration for the minimization of the function f and determine all initial points (x_0, y_0) such that the Gradient iteration with line search parameter $\alpha = 1$ yields $(x_1, y_1) = (1/3, 1/3)$. [4 marks]

e) Consider the Newton's iteration for the minimization of f. Show that selecting the initial points as $(1/9, 1/9)$, or $(7/9, 1/9)$, or $(1/9, 7/9)$ yields $x_1 = (1/3, 1/3)$. [2 marks]
(Hint: rewrite Newton's iteration as $\nabla^2 f(p_0)(p_1 - p_0) = -\nabla f(p_0)$, with $p_i = [x_i, y_i]^\top$.)

f) Suppose now that the decision variables x and y are such that

$$x \geq 0, \qquad y \geq 0, \qquad x+y-1 \leq 0.$$

Determine, without using necessary conditions of optimality, the global minimizer for the problem

$$\begin{aligned} &\min_{x,y} xy(x+y-1), \\ &x \geq 0, \\ &y \geq 0, \\ &x+y-1 \leq 0. \end{aligned}$$

[4 marks]

2. Consider the constrained optimization problem

$$\min_{x_1,\, x_2} \; x_1^2 + x_2^2$$
$$(x_1 - 1)^3 - x_2^2 = 0.$$

a) Plot on the (x_1, x_2) plane the level lines of the objective function and the admissible set. [4 marks]

b) Study regularity of the points in the admissible set. [2 marks]

c) Solve, graphically, the considered problem and determine the optimal cost. [2 marks]

d) Write the first order necessary conditions of optimality and show that these do not return any candidate optimal point. Explain why this is the case. [4 marks]

e) The constraint can be used to eliminate the term x_2^2 in the cost. Write explicitly the resulting problem and show that this problem does not have a solution. Explain why this is the case. [4 marks]

Show that a correct elimination of the variable x_2 yields a reduced problem in x_1 with an inequality constraint. Write the problem and determine its solution. Show that this coincides with the solution determined in part c). [4 marks]

3. Consider the optimization problem

$$\begin{aligned} &\min_{x_1,\, x_2} \; x_1 + x_2, \\ &x_1^2 + (x_2 - 1)^2 - 1 = 0, \\ &x_1^2 - x_2 = 0. \end{aligned}$$

a) By plotting the admissible set and the level lines of the objective function determine the optimal solution of the problem.
(Hint: the admissible set is composed of three isolated points.) [4 marks]

b) Argue that the considered problem has two global minimizers and determine regularity of these optimal points. [2 marks]

c) Write first order necessary conditions of optimality for the considered problem and determine candidate optimal solutions. Explain why only one of the global minimizers determined in part a) satisfies the necessary conditions of optimality. Let this candidate solution be the point P_1. [4 marks]

d) Write the augmented Lagrangian for the problem using the values of λ_1 and λ_2 associated to P_1. [2 marks]

Show that the augmented Lagrangian determined above has a stationary point at P_1 for any $\varepsilon > 0$. [4 marks]

Determine for which values of $\varepsilon > 0$ the augumented Lagrangian has a strict minimizer at P_1. [4 marks]

4. Consider the so-called economic dispatch problem with two generators, G_1 and G_2, and one load.

In what follows all *units* are omitted.

Let $P_1 \geq 0$ be the power produced by G_1 and $P_2 \geq 0$ be the power generated by G_2. Assume that the load demand is 10, that is $P_1 + P_2 = 10$, and that the cost of production for G_1 is

$$C_1(P_1) = 1 + 2P_1 + P_1^2,$$

whereas the cost of production for G_2 is

$$C_2(P_2) = 1 + 4P_2 + P_2^2.$$

Let the cost to be minimized be $C_1(P_1) + C_2(P_2)$ and suppose that, in addition to the constraint $P_1 + P_2 = 10$, P_1 and P_2 have to be such that

$$P_1\ P_2 \leq 16.$$

a) Sketch the admissible set and the level lines of the cost. Show that all points are regular points for the constraints. Determine, using your sketch, the solution of the problem. [6 marks]

b) Write first order necessary conditions of optimality for the considered problem. (Do not include the non-negativity constraints $P_1 \geq 0$ and $P_2 \geq 0$ in the formulation.)

[2 marks]

c) Using the conditions in part b) determine candidate optimal solutions.

[6 marks]

d) Explain why the second order sufficient conditions of optimality cannot be used to classify the candidate points determined in part c). [2 marks]

To determine the optimal solution eliminate the equality constraint and determine a reduced optimization problem with one inequality constraint. Solve the problem to determine the global minimizer of the considered economic dispatch problem. [4 marks]